Git & GitHub

Introduction:

Git is a version control system that tracks changes in files, while Github is a web-based platform that hosts Git repositories & provides collaboration features.

Git:

→ It is a free, open-source, distributed version control system (DVCS) used to track changes in source code & other files during software development.

Features:

1) Every developer can maintain a local copy of the repository & its history.

2) Records changes made to files over time

3) Allowes developers to work on diff. features independently.

4) Combines changes from different branches.

5) Supports sharing & integrating changes from multiple developers.

Terminologies

- Repository - a storage location containing project files + version history
- Working directory - The directory where files are viewed & modified.
- Staging Area - An area where selected changes are prepared for the next commit
- Commit - A recorded snapshot of the staged changes
- Branch - An independent line of developme
- Merge - process of combining changes from different branches.
- Clone - Creating a local copy of an existing repository
- Remote - A reference to a repository hosted elsewhere
- Push - send local files to remote repository
- pull - fetching remote changes & integrating them into current branch.
- Fetch - Downloading updates from a remote repository without automatically integrating the
- HEAD - It identifies the currently checked out commit / branch position.

Git Lifecycle

Working directory : Our project files are here. We can make changes

↳ git add

↳ Staging Area : We select which changes to include in next commit

↳ git commit

↳ Local Repository : Our changes are saved in our local .git repo.

↳ git push

↳ Remote Repository : Our commits are uploaded to a remote server (Github)

File States in Git

Untracked : A file is not tracked yet

↳ Unmodified : The tracked file has no changes

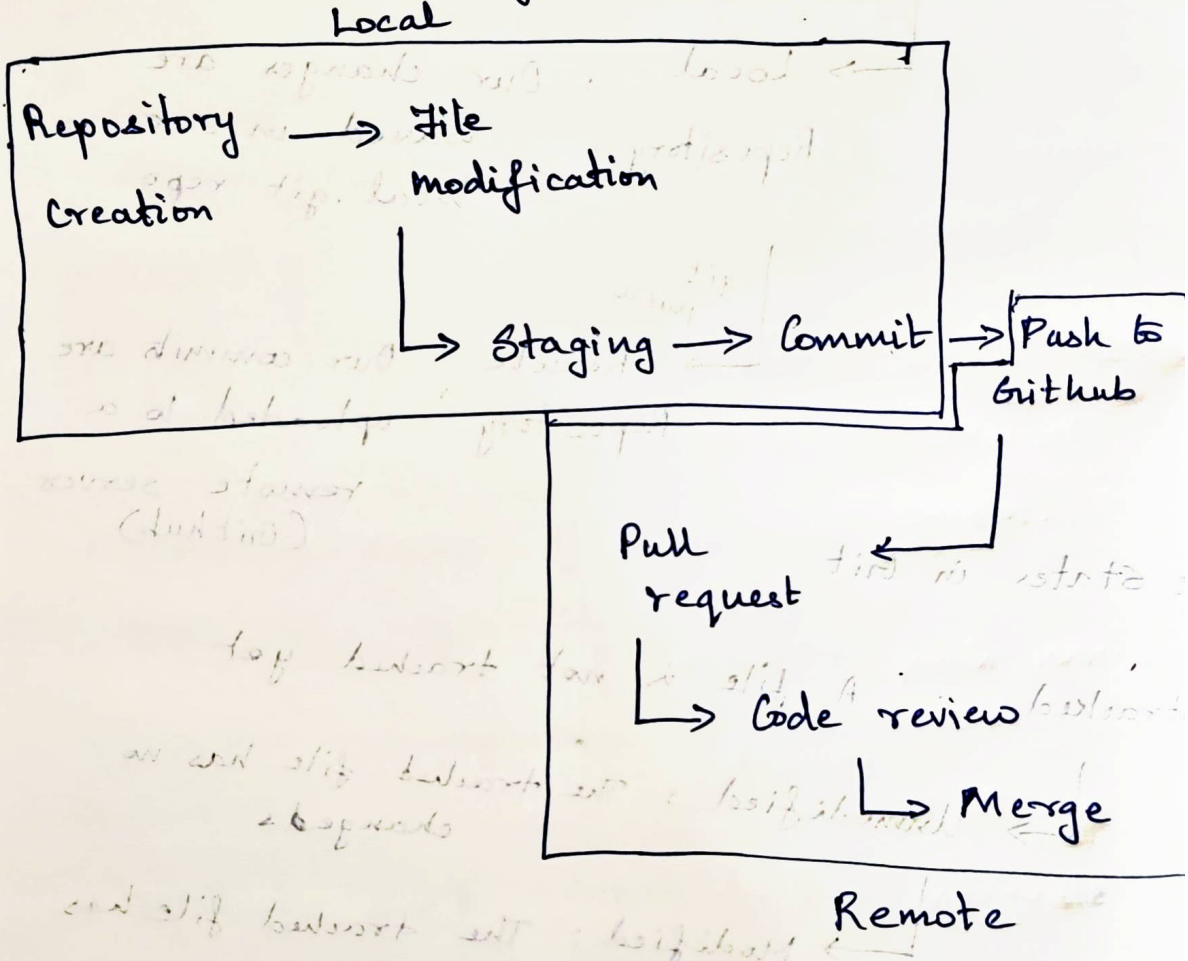
↳ Modified : The tracked file has changed.

↳ Staged : The file has been selected for next commit.

GitHub :

→ It is a cloud-based platform for hosting Git repositories. It provides tools for storing code online, collaborating with other developers, reviewing changes & so on.

Git & Github Workflow



Basic commands.

1) git --version : to check the git's version

2) git init : to start a new, empty repository in the current ~~.git~~ folder.

3) git status : It shows the changes done on a folder

4) mkdir <dir_name> : It creates a new, empty directory

5) cd <dir_name> : It moves from current to next/mentioned directory.

6) touch <filename.ext> : It creates a new, empty file.

7) ls : shows files & directories present in the current directory as a list.

8) git add <files> : adds a file to staging area

9) git add . : adds all files in the director to staging area

10) git commit -m <message> : It saves the changes in the current directory.

11) git log : view commit history

12) git log --oneline : view commit history in a single line.

13) git rm <file> : removes a tracked file

14) git mv <old><new> : renames/moves a tracked file.

15) git branch : list local branches

16) git branch <name> : create a branch

17) git switch <name> : switches branches

18) git switch -c <name> : create & switches a branch.

19) git branch -d <name> : deletes a merged branch

20) git remote -v : view remote repositories

21) git remote add origin <url> : Add a remote name 'origin'

22) git remote set-url origin <url> : change the URL of origin.

23) git push -u origin main : push main & set its upstream.

24) git pull origin main : fetch & integrate changes from main

25) git config --global user.name <Name> : set username

) git config --global user.email <mailid> : set email